

Java Container Awareness

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Introduction

Course Title: Java Container Awareness

Course description: This course provides a practical introduction to running Java applications as a fully containerized application, with a focus on resource detection and memory management inside a container. Learn how Java applications containerized applications integrate with Kubernetes and OpenShift and how to identify common runtime and resource-related issues. Explore essential troubleshooting techniques and tools for containerized Java workloads, particularly for Red Hat build of OpenJDK images, but it can be generalized.

Duration: 2 Hours

Topics covered in this course

- Introduction to Containerized Java
- Container Resource Detection
- Risks and Memory Management
- OpenShift and Kubernetes Integration
- Troubleshooting and Tooling
- Hands-on Lab

Prerequisites

This course assumes that you have the following prior experience:

- Experience with Java (in general) and heap (off-heap) allocation
- Experience with container orchestrators, such as Openshift in terms of deployment, not necessarily with Operators



Use of Artificial Intelligence: This publication contains content or media that is generated or assisted by AI tools. The Training deliverables have primarily human authorship, and all content was reviewed by an human domain expert.

Lab Environment



This lab requires a standard OpenShift Container Platform (OCP) environment with permissions to deploy an application using a StatefulSet, Pod YAML, or Deployment. You may use any OpenShift cluster you have access to. If you do not have access to an OpenShift cluster for the hands-on exercise, follow the instructions on this page to order a lab environment.

Instructions to Launch Your Lab on the Red Hat Demo Platform (RHDP)

1. **Log in** to the [RHDP portal](#)
2. **Search** for the catalog entry: **Red Hat OpenShift Container Platform Cluster (Multi-Cloud)**.
3. **Click** on the catalog entry in the search results.
4. On the catalog page, **click** the **Order** button.
5. **Fill out** the required details in the order form.
 - **Activity:** *Practice / Enablement*
 - **Purpose:** *Learning about the product*
 - **Cloud Provider:** *cnv*
 - **OpenShift Version:** *4.21 (or as desired)*
 - **cluster size:** *sno*
6. **Review the warning** at the bottom of the form and **check the box** labeled: **“I confirm that I understand the above warnings.”**
7. **Click** the **Order** button to place your lab order.

Important Notes:

- This lab may take approximately **5 minutes** to become ready.
- You will receive an **email with access details** once your lab environment is ready.
- You can also **retrieve lab access** directly from the RHDP portal.

How to Access Your Lab via the RHDP Portal:

1. On the RHDP portal, **click** on the **Services** option in the left-hand menu.
2. **Select your lab** from the listings on the right-hand side of the page to view access details.

RHDP Portal Links

- RedHat associates: <https://demo.redhat.com/>
- RedHat partners: <https://partner.demo.redhat.com/>

Introduction to Containerized Java

Containerized Java focuses on the evolution of the Java Virtual Machine (JVM) to operate bounded and efficiently within container environments. Therefore this will impact deployments in orchestrators such as OpenShift and Kubernetes. Currently, Java runtimes utilize container awareness to detect cgroup limits, allowing the JVM to automatically adjust its resource restriction based on container boundaries rather than host hardware limits.

Key aspects include:

- **Evolution of Java:** Shifting from host-based resource perception to environment-aware detection.
- **Container Awareness:** The ability of the JVM to recognize and respect memory and CPU limits imposed by the container runtime.
- **Cgroup Detection:** Leveraging kernel-level control groups to monitor and manage resource availability accurately.
- **Operational Efficiency:** Simplifying deployments by ensuring the JVM aligns its internal memory and thread management with defined container resource limits.
- **Red Hat build of OpenJDK:** Red Hat build of OpenJDK containers, have specific settings defined in the `run-java.sh` script, which is the default **ENTRYPOINT** when using our image and adds features and configurations.

This course is fully based on the current OpenJDK implementation and the article [How to use Java container awareness in OpenShift 4](#).

Evolution of Java in containerized environments

The landscape of Java application deployment has shifted dramatically with the advent of containerization. Initially designed for virtual machines and bare-metal hardware, the Java Virtual Machine (JVM) faced significant architectural hurdles when introduced to the world of Linux containers.

Non-container-aware Java

Before 2019, Java (its JVM) would identify the system resources by querying the underlying operating system (OS) kernel via `/proc/meminfo` and `/proc/stat`. When a Java process runs on a physical server, it sees the total memory and CPU cores of the host.

In other words, despite running inside a container, Java was unable to detect the container's limits. It would only detect the host limits regardless of the deployment. Consequently, the DevOps team needed to set the heap limits (Xmx) manually as a size and make sure those two were appropriate: container size and Java heap size.

Later, evolving this concept, Java was encapsulated inside an initialization script: `run-java.sh`, which would then detect the container limits and deploy Java with those limits (script → container limits → Java). Java did not know about cgroups. It would just start, and the initialization

script would provide its max heap, Xmx, and CPU limits.

JDK-8146115

The evolution toward true container awareness reached a milestone with **JDK8146115**. This critical enhancement, backported to Java 8 (starting with 8u191) and native to all newer versions (Java 11, 17, 21+), fundamentally changed how the JVM initializes.

Instead of looking at the host's total resources, the JVM was updated to inspect the cgroup hierarchy directly. This allows the JVM to:

1. **Read Container Memory Limits:** Automatically calculate appropriate initial and maximum heap sizes (`-Xmx` and `-Xms`) based on the specific container memory constraints.
2. **Detect CPU Availability:** Properly identify the number of CPU shares or quotas allocated to the container, allowing the garbage collector and internal thread pools to scale their concurrency appropriately.

Containers achieve isolation through **Control Groups (cgroups)**, which is a Linux kernel feature that limits, accounts for, and isolates the resource usage of a collection of processes. Before 2019, Java was not cgroup-aware when running inside a container, so the JVM would still attempt to query the host's total resources rather than the container's restricted limits. This created a mismatch where the JVM believed it had access to gigabytes of host RAM, leading to:

- **OOMKills:** The JVM would attempt to allocate heap memory based on host capacity, triggering the container orchestrator to kill the process for exceeding its cgroup memory limit.
- **Performance Degradation:** The JVM would size its internal thread pools (e.g., ForkJoinPool) based on the total physical CPU cores of the host, leading to excessive context switching and performance bottlenecks in environments where the container was throttled, given the container likely has different CPU limits.

Current Fully Container-Aware Java

Today, container awareness is the standard expectation for cloud-native Java development for Java 8u372+, 11.0.16+, 17+ such as 21 and 25. The transition from unaware to fully container-aware has considerably streamlined the DevOps workflow:

- **Self-Tuning:** Applications can now adjust their memory footprint automatically to fit within Kubernetes/OpenShift resource requests and limits upon start/restart.
- **Improved Stability:** By aligning the JVM's view of resources with the kernel-enforced limits, the frequency of container restarts due to resource exhaustion is drastically reduced.
- **Reduced Configuration Overhead:** Developers do not spend time tuning CPU values and heap limits (`-Xmx`), as the JVM can now effectively derive them from the container environment deployed by an orchestrator via percentage automatically by having `MaxRAMPercentage` at 25% (upstream value).

Specifically in terms of Red Hat build of OpenJDK, our images rely on the script `run-java.sh` script, which current is not use anymore for detecting limits from the container but to add features: such as `JAVA_OPTS` and `JAVA_OPTS_APPEND` environment variables and configurations

(MaxRAMPercentage at 80%).

Understanding this evolution is important for any team moving toward microservices architectures. Container awareness provides a robust foundation that requires the development team (who handles the Java settings) and the Deployment team (who handles the container size) to balance between JVM heap memory, off-heap memory, and the container's hard limits, which is a topic that remains a pivotal consideration for production-grade containerized deployments.

Overview of container-aware versus non-container-aware Java

Before containerization and container awareness, Java applications more often than not would struggled to coexist within its container limits. This friction stemmed from a fundamental mismatch: the Java Virtual Machine (JVM) was designed to query the host operating system for resource availability, while containers rely on control groups (cgroups) to impose different resource constraints.

The Evolution of Container Awareness

Before the introduction of container awareness, the JVM would query the host machine's total physical memory and CPU cores, completely ignoring the limits imposed by the container runtime.

Feature	Non-Container-Aware (Legacy)	Fully Container-Aware (Current)
Memory Detection	Sees total host RAM	Sees container (cgroup) defined memory limit
CPU Detection	Sees total host CPU cores	Sees container (cgroup) defined CPU limit
Heap Sizing	Based on total host memory	Based on container memory limit
OOM Risk	High (dissociated with the container size)	Lower (associated with the container size)

Container awareness is the default (already enabled) and the user needs to manually set the flag `-UseContainerSupport` when starting Java to disable it. That is to say: the default is `+UseContainerSupport`, so Java is container aware by default: if you do not set this flag, OpenJDK will be container aware.

Non-Container-Aware Java: The Visibility Gap

In a non-container-aware environment, if you deploy a Java application with a default heap size (e.g., `-Xmx` not explicitly defined) into a container with a 1GB limit, the JVM will see the host's 64GB of RAM and attempt to allocate a heap significantly larger than 1GB, causing issues.

When the JVM allocates this memory, the host kernel will trigger a cgroup **Out-of-Memory (OOM) Kill** event. The process is terminated abruptly because it breached the container's allowed

boundary, despite the JVM assuming it had plenty of memory available. Note that JVM allocation and application consumption are separated and may considerably diverge, because the JVM may allocate more than the application is in fact currently using.

Container-Aware Java: The Shift with JDK-8146115

The industry shift occurred with the implementation of **JDK-8146115**, starting with Java 8u191 and then other subsequent jiras on all current LTS releases (e.g., Java 11, 17, 21, 25) that now rely on cgroups natively. In other words, those updates enabled the JVM to detect its container **cgroup** filesystem directly natively.

When running a container-aware JVM, the process performs the following:

- **Memory Sensing:** Instead of asking the OS for total physical RAM, the JVM reads the **memory.limit_in_bytes** file from the container's cgroup. It then calculates the default heap size as a percentage of this limit, rather than a percentage of the host's total RAM.
- **CPU Awareness:** The JVM queries the **cpu.shares** or **cpu.cfs_quota_us** files to determine how many CPU cores are available to the container. And calculates the thread pools from this value, rather than the host limits.

Why This Matters for current Java Deployments

Transitioning to container-aware Java is not merely a technical optimization; it is a prerequisite for stability in orchestrators like OpenShift and Kubernetes.

- **Simplified Configuration:** Developers no longer need to hardcode **-Xmx** values that perfectly match every environment's resource limit. The JVM automatically adapts to the **limits** defined from the container/cgroups limits from the deployment manifests and uses percentage as reference.
- **Deterministic Behavior:** By aligning the JVM's view of resources with the container's reality, you reduce the likelihood OOM kills that occur when the JVM allocates memory, given it won't over-allocate (above the container size).
- **Operational Efficiency:** Container-aware Java allows DevOps teams to set resource requests and limits in OpenShift with the confidence that the application will scale its internal memory management to fit the provided space.

The default value of **MaxRAMPercentage** is 25% for upstream deployment, however Red Hat build of OpenJDK will allow taking advantage of **run-java.sh** will have the following impacts:

*The JVM achieves container awareness (memory and CPU). * The heap will be calculated at 50% of the container size (not the host). Later images will default to 80%. *The user can use **JAVA_OPTS** and other environment variables from **run-java.sh**, such as **JAVA_OPTS**, **JAVA_OPTS_APPEND**, **GC_MAX_METASPACE_SIZE**.



While container awareness resolves around JVM limits association with the container size, such as memory (heap) and CPU (for thread calculation). It does not necessarily account for full memory utilization, especially **off-heap** memory (e.g., metaspace, code cache, direct buffers, or native memory). Therefore, even with

container aware Java, one must benchmark your application to ensure that the ratio of heap to off-heap memory is sufficient resources are allocated to avoid OOMkills events

Importance of cgroup detection for deployments

In cloud-native architectures, the shift from virtual machines to containerized runtimes has fundamentally changed how applications interact with hardware. For Java, historically accustomed to seeing the full host OS settings, the introduction of cgroups (control groups) represents a change and evolution in its resource detection and management.

The Evolution of Java and Resource Visibility

Before container-aware JVMs, a Java application running inside a container would often query the underlying host OS to determine available memory and CPU. If the container was restricted to 2GB of memory, but the host node had 64GB, the JVM would frequently miscalculate its maximum heap size (`-Xmx`). This led to two primary failure modes:

1. **OOMKiller Invocation:** The JVM, believing it had access to a larger pool of memory, would allocate heap space that exceeded the container's hard limit. The Linux kernel, seeing the container exceed its cgroup limit, would trigger an Out-of-Memory (OOM) Kill, abruptly terminating the process.
2. **Resource Inefficiency:** Conversely, if the JVM underestimated the capacity, it might fail to utilize the resources explicitly allocated to it by the orchestrator, leading to poor performance and degraded throughput.

Why cgroup Detection is Critical

Cgroup detection is the mechanism by which the JVM detects the boundaries imposed by the container runtime. Its importance for deployments cannot be overstated for the following reasons:

- **Boundary Compliance:** It ensures the JVM respects the container's quota rather than the physical host's capacity. By reading from `/sys/fs/cgroup/` (or equivalent cgroup v2 paths), the JVM adjusts its internal ergonomics to fit within the `cgroup` limits.
- **Preventing Failures:** With proper cgroup detection, the JVM aligns its garbage collection triggers and heap allocation thresholds with the container's memory limit. This prevents the scenario where the application performs perfectly under light load but crashes instantly when the memory pressure hits the cgroup ceiling.
- **Orchestration Alignment:** Orchestrators like OpenShift and Kubernetes use cgroups to enforce Resource Limits and Requests. Java's ability to read these limits allows it to operate nominally inside a container. When the JVM is container-aware, it will effectively use flags like `-XX:MaxRAMPercentage`, ensuring the heap scales dynamically with the pod's assigned limit when it is deployed. Users can tune that value.

Implications for DevOps teams

For a DevOps team, relying on container-aware Java simplifies the deployment pipeline. You no

longer need to hardcode `-Xmx` values for every environment. By allowing the JVM to detect the cgroup limit, you can:

- **Standardize Images:** Use the same container image across development, testing, and production environments, where resource limits may vary significantly.
- **Predictable Scaling:** As you adjust memory limits in your Kubernetes manifest, the application automatically adapts its memory footprint without requiring a rebuild or manual configuration update.

If before the users set `Xmx` and the container size, currently they just need to know the adequate percentage and the heap will adapt to the container size automatically at start up.

Red Hat build of OpenJDK containers

Specifically the default OpenJDK container image, the default ENTRYPOINT will be `run-java.sh`, which is a script that provides default settings, such as `ParallelGC` as expected GC, but also environment variables that can be used at runtime:

Environment Variable	Purpose
<code>JAVA_OPTS</code>	Main option, which includes default settings - can be used to replace them
<code>JAVA_OPTS_APPEND</code>	Append option, can be used to complement the main ones
<code>GC_COLLECTOR</code>	Used for GC collector

`JAVA_TOOL_OPTIONS` is not from the `run-java.sh` and should be avoided, it will be used for any Java environment variable, including `java` itself:

Example: Why `JAVA_TOOL_OPTIONS` should be avoided

```
$ export JAVA_TOOL_OPTIONS="EXAMPLE"
$ openjdk/app-injection$ java
Picked up JAVA_TOOL_OPTIONS: EXAMPLE
Unrecognized option: EXAMPLE
Error: Could not create the Java Virtual Machine.
Error: A fatal exception has occurred. Program will exit.
```

For a deep dive into how these changes specifically affect Java runtimes specifically for cgroups v2, review the scenarios outlined in the following developer article:

- [How does cgroups v2 impact Java, .NET, and Node.js in OpenShift 4? - Scenarios with Java](#)

Key Takeaways for Java Developers - OpenJDK 8u372+, 11.0.16+, 17, and 21 natively support cgroups v2 detection. - Red Hat container images now rely on native OpenJDK C++ container detection rather than legacy `run-java.sh` wrapper scripts for setting memory limits. - The `MaxRAMPercentage` property (defaulting to 80% in these images) allows the JVM to dynamically size its heap based on the cgroups v2 boundaries.



While cgroup detection prevents the JVM from over-allocating heap space by setting the Heap Percentage automatically and allowing the user to tune it, it does **not** manage off-heap memory (Metaspace, Code Cache, Thread stacks, etc.). Because the heap is only a portion of the total memory usage, administrators must still account for the non-heap overhead separately. Failing to leave enough room for off-heap memory can still result in a cgroup OOMKill, even if the heap appears to be within the allowed percentage.

By ensuring your deployments utilize container-aware Java versions (such as Java 8u191+ or any later LTS version), you bridge the gap between the application's runtime needs and the platform's enforcement mechanisms, leading to more resilient and stable microservices. The user only needs to worry about the exact percentage to deploy the application.

Container Resource Detection

Container resource detection enables the Java Virtual Machine (JVM) to accurately identify resource constraints imposed by the container environment rather than the underlying host node.

- **JVM Evolution:** Starting with Java 8u191 (via JDK8146115), the JVM gained the ability to parse control group (cgroup) files to determine resource boundaries.
- **Memory Limit Detection:** The JVM identifies the specific memory limit assigned to the container, allowing it to scale the heap size appropriately based on available container resources.
- **CPU Availability:** The JVM detects the number of CPU cores allocated to the container, preventing over-provisioning and ensuring efficient thread management within the containerized environment.
- **Operational Benefits:** Accurate detection prevents OutOfMemory and OutOfMemoryKill caused by the JVM incorrectly defaulting to host-level physical memory rather than container-specific limits.

In simple terms: don't not worry about the Xmx directly. Worry about the correct percentage value to be set of heap vs off-heap.

Understanding JDK8146115 and Java 8u191 Updates

Before the introduction of specific container-aware updates, the Java Virtual Machine (JVM) lacked the ability to natively read the resource boundaries imposed by container runtimes. This caused significant operational friction in cloud-native environments.

The Challenge: JVM Resource Blindness

In early containerized environments, the JVM would query the underlying host operating system for available resources (such as total system memory and available CPU cores) rather than the specific cgroup limits applied to the container.

This behavior led to a critical disconnect:

- **Memory Miscalculation:** The JVM would attempt to size its heap based on the host's physical RAM, often exceeding the container's hard memory limit set by Kubernetes or OpenShift.
- **OOM Killer Intervention:** Once the Java process exceeded the container's cgroup memory limit, the OCP node kernel's Out-Of-Memory (OOM) Killer would terminate the process, leading to unexpected pod restarts.
- **CPU Throttling:** Similar issues occurred with CPU calculation, where the JVM would perceive all host cores as available, leading to inefficient thread pool sizing and potential overhead, such as context-switching and multiple threads competing for resources.

The Turning Point: JDK8146115

To address these issues, the OpenJDK community introduced **JDK8146115**. This enhancement

provided the necessary infrastructure for the JVM to correctly read and respect cgroup memory and CPU limits.

Instead of querying `/proc/meminfo` or `/proc/stat`—which reflect host-level resources—the JVM was updated to inspect the cgroup filesystem (`/sys/fs/cgroup/`) directly. This allows the JVM to identify the specific constraints applied to the process namespace.

Java 8u191: The Paradigm Shift

While JDK8146115 provided the foundational logic, the release of **Java 8u191** served as the catalyst for widespread adoption. This update backported the container-awareness features, effectively bringing modern cloud-native capabilities to long-standing Java 8 applications.

Key changes introduced in 8u191 include:

- **Native Cgroup Awareness:** The JVM now automatically detects container limits at startup.
- **Default Ergonomics:** The default heap size calculation logic was updated to use a fraction of the container’s memory limit instead of the host’s physical memory.
- **New JVM Flags:** Introduction of flags such as `-XX:MaxRAMPercentage`, `-XX:MinRAMPercentage`, and `-XX:InitialRAMPercentage`, allowing developers to define heap sizes as a ratio of the container limit rather than hard-coded values.

Impact on Modern Deployments

With Java 8u191+, the relationship between the container and the JVM became **coupled**.

Feature	Legacy Behavior (Pre-8u191)
Modern Behavior (8u191+)	Memory Source
Queries <code>/proc/meminfo</code> (Host RAM)	Queries cgroups (Container Limit)
Heap Sizing	Requires manual <code>-Xmx</code> configuration
Automatically scales based on <code>-XX:MaxRAMPercentage</code>	Operational Impact
Decoupled; changing container size requires manual JVM tuning	Coupled; scaling container limits automatically adjusts heap

By aligning the JVM’s view of resources with the container orchestration layer, these updates eliminated the decoupling problem (container not matching the Java’s heap size). Developers no longer need to manually synchronize `-Xmx` settings with pod resource limits, significantly simplifying the management of Java applications in Red Hat OpenShift and Kubernetes.

Detecting Container Memory Limits

In modern Java deployments, it is critical that the Java Virtual Machine (JVM) correctly identifies the memory boundaries imposed by the container orchestration platform. Failure to do so leads to performance degradation, inefficient resource utilization, or unexpected pod terminations.

Why Container Awareness Matters

Prior to the integration of `JDK8146115` (introduced in Java 8u191), the JVM relied on standard OS-level utilities to detect available memory. In a containerized environment, these utilities often reported the total memory of the underlying physical host (the OCP node) rather than the restricted memory assigned to the specific container.

If the JVM perceives the node's total memory instead of the container's limit, it may attempt to allocate a heap size that exceeds the container's allowed threshold, resulting in an `OutOfMemoryException` or, more commonly, a `SIGKILL` by the Linux kernel's Out-Of-Memory (OOM) killer.

How the JVM Detects Container Limits

When running in a container-aware environment, the JVM inspects the control group (`cgroup`) filesystem. By reading the files exposed within the container—specifically those under `/sys/fs/cgroup/memory` - the JVM determines the hard limits imposed by the container orchestrator.

Verification Tools

To determine if your Java application is correctly "seeing" the container limits, you should utilize built-in JVM diagnostic flags rather than relying on standard Linux utilities like `free -m`.



Standard utilities like `free`, `top`, or `lscpu` are **not container-aware**. They will report the resources of the host node, leading to incorrect diagnostic conclusions. Use container-aware tools, such as `jcmd VM.info`, and `-XshowSettings:system`.

Using `-XshowSettings:system`

The most direct way to verify what the JVM has detected is to pass the `-XshowSettings:system` flag during application startup or via `jcmd` instrumentation at runtime:

Example Output:

```
Operating System Metrics:
  Provider: cgroupv1
  Memory Limit: 128.00M
  Memory Soft Limit: unlimited
  Memory Current: 45.21M
  CPU Shares: 512
  CPU Quota: 100000
  CPU Period: 100000
```

When the JVM is container-aware, you will see explicit lines reflecting the `Memory Limit` and active CPU.

Consequently:

- If the memory value matches your OpenShift/Kubernetes pod YAML limit configuration, the JVM is successfully performing container detection.
- If the CPU value matches your OpenShift/Kubernetes pod YAML limit configuration at 25% so then `run-java.sh` is not being

used and another ENTRYPOINT is being used. Otherwise, if matches 80% (or 50%) likely run-java.sh is being used.

Troubleshooting Memory Misalignment

If your application exhibits the following symptoms, the JVM is likely failing to respect container limits:

1. **Host-level reporting:** Non-container-aware commands executed inside the container (e.g., `free -m`) show memory values matching the physical node rather than the pod limit.
2. **OOM Killer termination:** The pod restarts with an exit code indicating OOM, despite the `-Xmx` setting appearing reasonable for the application's perceived workload.

Two main scenarios:

1. **Underutilization:** The application heap is significantly smaller than the container limit because it defaulted to a generic fraction of the host's massive memory footprint, leading to frequent Garbage Collection (GC) pauses.
2. **Overutilization:** The application heap (plus off-heap) is significantly larger than the container limit, which will eventually lead:
 - To OOMKill events
 - To OOMException events
 - SIGTERM, in case the JVM is not responsive

Diagnostic Strategy

To analyze resource constraints effectively:

- **Check the logs:** Inspect your pod logs to confirm if `run-java.sh` or the standard OpenJDK startup scripts are being utilized, as these often contain logic to calculate heap sizes based on container limits.
- **Verify jcmd output:** Use `jcmd <pid> VM.info` to inspect the runtime configuration and verify `cgroup` memory controller details.
- **Compare with deployment Metadata:** Always compare the JVM's reported `Memory Limit` against the `resources.limits.memory` configured in your OpenShift `DeploymentConfig` or `Deployment` YAML and CPU with the active CPUs.

Determining CPU Availability Within Containers

In a virtualized or bare-metal environment, the Java Virtual Machine (JVM) traditionally queries the operating system for the total number of physical CPUs available on the host. However, in containerized environments like Red Hat OpenShift or Kubernetes, a container is typically restricted to a subset of the host's compute resources.

If the JVM is not container-aware, it may incorrectly identify the host's total CPU count rather than the container's allocated limit. This mismatch can lead to inefficient thread pool sizing (such as

`ForkJoinPool` or parallel garbage collection threads), causing contention and performance degradation.

How Java Detects CPU Resources

Current JDKs (Java 8u191+ and all subsequent versions) use **cgroups (Control Groups)** to determine CPU availability. The JVM reads the cgroup filesystem to calculate the `ActiveProcessorCount`.

- For cgroups v1, versions Java 8u191+ should be enough
- For cgroups v2, versions Java 8u372+ Java 11.0.16+ are required.

The Calculation Logic

When running inside a container, the JVM inspects two primary metrics from the cgroup controller to determine how many processors it should use:

1. **CPU Shares (`cpu.shares`):** This represents the relative weight of the container. If the host has 64 cores and your container has a share weight that equates to 2 cores, the JVM uses this to influence its internal parallelism.
2. **CPU Quota and Period (`cpu.cfs_quota_us` and `cpu.cfs_period_us`):** These define a hard limit. For example, if the quota is 200,000 and the period is 100,000, the container is limited to 2 cores ($200,000 / 100,000 = 2$).

The JVM calculates the number of processors as the minimum of the host's processors and the calculated container limit.

Diagnostic Tools: Container-Aware vs. Non-Aware

To verify how your Java application perceives CPU availability, you must distinguish between tools that see the "Host" view versus the "Container" view.

Tool Category	Examples
Insight	Non-Container Aware
<code>lscpu</code> , <code>nproc</code>	Returns the total CPU count of the underlying OpenShift Node . This is often misleading for capacity planning.
Container Aware	<code>VM.info</code> , <code>-XX:+PrintFlagsFinal</code>

Verifying CPU Detection

You can inspect the JVM's view of your container's resources by using the `jcmd` utility.

1. Access your pod terminal: `oc rsh <pod-name>`
2. Run `jcmd` to print VM information: `jcmd <pid> VM.info`

Look for the `container` section in the output. A container-aware JVM will report output similar to this:

```
container (cgroup) information:
container_type: cgroupv1
cpu_cpuset_cpus: 0-127
cpu_memory_nodes: 0-7
active_processor_count: 1 <-- JVM correctly sees 1 CPU limit
cpu_quota: -1
cpu_period: 100000
cpu_shares: 51
```

Best Practices for CPU Configuration

- **Define Requests and Limits:** Always explicitly define `resources.limits.cpu` in your OpenShift deployment YAML. This ensures the cgroup values are populated correctly.
- **Avoid Oversizing:** If you set a high CPU limit but your application is single-threaded, the JVM may spawn excessive GC threads, leading to context-switching overhead.
- **Monitor `active_processor_count`:** If `active_processor_count` does not match your expectations, ensure you are running a supported OpenJDK version (8u191 or higher). If you are forced to use an older, non-aware version, you must manually set `-XX:ActiveProcessorCount` to match your container limit.

Hands-on Exercise: Inspecting CPU Awareness

In this short task, you will verify the JVM's CPU detection.

1. Create new project:

Use the following command to create a new OpenShift project:

```
oc new-project my-java-app
```

2. Launch the container:

Use the following command to launch a new container:

```
oc create deployment my-java-app --image=quay.io/hummingbird/openjdk:latest
```

3. Connect to the container:

Use the following command to connect to your running container:

```
oc get pods
oc rsh <pod-name>
```

4. Inspect current CPU perception:

Inside your running container, identify the Java process ID (use `jcmd`, `jps`, or `ps aux`):

```
jps
```

Example output:

```
sh-5.3$ jps
1 JShellToolProvider
84 Jps
27 RemoteExecutionControl
sh-5.3$
```

5. Query the VM metadata:

Run the following command to see the active processor count:

```
jcmd 1 VM.info | grep active_processor_count
```

Example output:

```
sh-5.3$ jcmd 1 VM.info | grep active_processor_count
active_processor_count: 32
sh-5.3$
```

6. Compare with OS-level tools:

Run `nproc` in the same terminal.

```
nproc
```

Example output:

```
sh-5.3$ nproc
32
sh-5.3$
```



If you deploy a container with limits (not just requests, but limits) and `active_processor_count` (`jcmd` or `java -XshowSettings:system`) returns that limit, the application is correctly container-aware. Otherwise, in case it returns the host values, this indicates the application is likely not container aware (`nproc` shows host).

7. Set the resource limits on the deployment.

```
oc set resources deployment/my-java-app --requests=cpu=1000m --limits=cpu=2000m
```

8. Connect to the newly running container.

```
oc get pods  
oc rsh <pod-name>
```

9. Verify the CPU limits honored by the container and the Java application running inside it.

```
nproc  
jcmd 1 VM.info | grep active_processor_count
```

Example output:

```
sh-5.3$ nproc  
1  
sh-5.3$ jcmd 1 VM.info | grep active_processor_count  
active_processor_count: 1  
sh-5.3$
```

Notes: Given the fact only integers can be used as CPU boundaries:

- Values below 1 CPU, are rounded to 1.
- Broken values are rounded to the lowest integer, so 2200 millicores, becomes 2 CPU.

Risks and Memory Management

When deploying Java in containerized environments, improper memory configuration can lead to significant stability and performance issues. Understanding the balance between heap and off-heap memory is critical to preventing infrastructure-level failures.

- **Decoupling Risks:** Hard-coding heap sizes (e.g., via `-Xmx`) without considering container limits can lead to memory mismatches. If the JVM ignores container boundaries, it may trigger cgroups OOM-Kills, causing the kernel to terminate the application.
- **Off-Heap Considerations:** Container-aware Java calculates heap size as a percentage of available container memory. It is essential to account for off-heap usage (e.g., metaspace, thread stacks, native buffers) to ensure the total memory footprint does not exceed the container limit.
- **Resource Underutilization:** Inadequate configuration can lead to inefficient resource usage, where the JVM reserves significantly less memory than the container provides, leaving allocated infrastructure resources idle.
- **Non-container aware allocation:** Allocating a certain percentage for container, but not benchmarking/tracking non-container aware components inside the Java application. For example, jemallo and glibc's malloc that are below the JVM and are both not container aware.

Best Practices:

- Prioritize container-aware JVM settings over manual heap overrides to allow for dynamic scaling.
- Benchmark application memory usage to determine the optimal ratio between heap and off-heap requirements.
- Avoid configurations that ignore cgroup limits, as these bypass the self-tuning capabilities of modern JDKs.
- Track specific percentage utilization for the application.
- Use limits in the container settings to make sure Java won't use host details.
- Understand Quality of Service (QoS) and how it impacts (Java) deployments.

Implications of Decoupling Heap Memory from Container Limits

Inside a containerized environment, the JVM's ability to detect container resource boundaries is a critical feature. When developers manually specify heap sizes using the traditional heap limit: `-Xmx` without regard for the container's cgroup limits, they risk decoupling the JVM memory footprint from the orchestration layer. This decoupling often leads to significant operational instability and an inherited risk of OOMKill.

The Risks of Manual Overrides

Before Java became fully container-aware, it was common practice to explicitly define `-Xmx` (maximum heap size) to ensure the JVM did not attempt to consume all available memory on a

physical host. For example, on a straight VM deployment. However, in a containerized world, this approach creates several hazards:

- **cgroup OOMKills:** If the `-Xmx` is set too high relative to the container's memory limit, the total memory usage—heap plus off-heap (metaspace, thread stacks, GC structures, and native buffers)—will eventually exceed the cgroup limit. The container runtime or the kernel will detect this violation and trigger an `OutOfMemoryKill` (OOMKill), terminating the process abruptly.
- **Resource Underutilization:** If the `-Xmx` is set significantly lower than the container limit, the application cannot scale its memory usage to handle increased load, even if the container has plenty of headroom. This results in inefficient resource allocation and potential `OutOfMemoryException` (OOMException) errors within the JVM, despite the container appearing to have available memory.
- **Off-heap limits:** DevOps teams/developers may often focus exclusively on the heap. However, native memory (off-heap) usage is not governed by `Xmx`, although `Xmx` caps the heap limit. The JVM is a combination of both heap and off-heap, and both segments together potentially cause the combined memory pressure to breach the container's hard limit.

Build the container with Xmx (manual settings is an Anti-Pattern)

Building an application with JVM settings hard coded is an anti-pattern. That's because a change on the JVM flag would require a new build, which breaks the main idea of Kubernetes.

In other words, baking JVM configurations directly into an application artifact is an anti-pattern that severely limits the flexibility of the application directly and impacts its deployment.

For Red Hat's OpenJDK containers, the `run-java.sh` allows `JAVA_OPTS` and `JAVA_OPTS_APPEND` environment variables that allow the user to set JVM flags, including MaxRAM Percentage and Garbage collector.

Decoupling is an Anti-Pattern

Decoupling occurs when the JVM's memory configuration ignores the orchestrator's resource constraints. This makes the application blind to the environment it resides in.

Modern Java versions (Java 8u191+, Java 11+) are designed to be container-aware, meaning they automatically derive memory settings from cgroups. Overriding this by hardcoding heap values removes the benefits of this feature, essentially reverting the JVM to a legacy mode that is non-container aware of its surroundings.

Best Practices: The Percentage-Based Approach

To avoid decoupling, the recommendation is to shift from absolute values (`-Xmx`) to relative values. The percentage already comes by default and the user can tune that via `-XX:MaxRAMPercentage`, the JVM maintains a coupled relationship with the container:

- **Dynamic Changes:** As you increase the memory limit of your OpenShift or Kubernetes pod, the JVM heap will not automatically adjust to maintain the defined percentage. That's because the JVM only detects the limits once when it starts and therefore needs to fully restart to detect its

limits, so pod restart/JVM start are required.

- **Safety Margin:** Leaving a percentage of the container memory free (i.e., not allocating 100% to the heap) provides the necessary space for off-heap components—such as Netty buffers, thread stacks, and Metaspace—preventing cgroup-level OOMKills.



If you must explicitly define memory, use `-XX:MaxRAMPercentage=N`. This ensures the heap size is always a derivative of the container limit, preserving the coupling between the JVM and the infrastructure.

Summary of Implications

Risk Factor	Consequence	Impact
Fixed <code>-Xmx</code>	Decoupling from cgroups	Unexpected OOMKills
Ignored Off-Heap	Native memory starvation	JVM instability / Crashes
Manual Constraints	Operational rigidity	Need for manual tuning during scaling

By allowing the JVM to detect container limits automatically, i.e. by utilizing percentage-based configurations, the DevOps teams ensure the Java application remains dynamic, resilient, portable, and correctly scaled within your cluster limits and boundaries.

Managing Off-Heap Memory in Containerized Environments

In current Java applications (Java 8+), memory management extends beyond the Java Heap. While the Heap is responsible for storing object instances, the **Off-Heap** memory—encompassing Metaspace, Code Cache, Thread stacks, and Direct ByteBuffers. Those are critical for the stability of containerized deployments.

When a Java application runs in a container, it is subject to strict Linux Control Group (cgroup) memory limits. If the sum of the Heap and Off-Heap memory exceeds the container limit, the container runtime will trigger an **OOMKill** (Out of Memory Kill).

The Decoupling Risk

Historically, developers focused exclusively on `-Xmx` (maximum heap size). However, in a container, the JVM sees the container's memory limit. If you configure the Heap to consume too large a percentage of the container's limit, you leave insufficient headroom for Off-Heap requirements. Therefore a percentage approach is used:

Key Off-Heap components that must be accounted for include:

- **Metaspace:** Stores class metadata.
- **Code Cache:** Stores compiled native code by the JIT compiler.
- **Thread Stacks:** Each thread requires native memory; in applications with high thread counts,

this can lead to significant memory consumption.

- **Direct Buffers:** Used by NIO libraries and netty-based frameworks.

Best Practices

To prevent **OOMKills**, you must treat the container's memory limit as the total capacity for both Heap and Off-Heap.

1. **Calculate the Total Footprint:** Before setting heap percentages, perform load testing to determine the baseline Off-Heap usage of your application under peak load.
2. **Use Relative Sizing:** Instead of fixed **-Xmx** values, JVM will already start with a percentage value that leverage container-aware JVM flags that allow the heap to scale proportionally to the container limit. Therefore, user can tune that value:

```
-XX:MaxRAMPercentage=70.0
```

Note: If your application is highly dependent on Off-Heap (e.g., uses large Direct Buffers or high thread counts), you should reduce this percentage to 50% or 60% to ensure enough spare space/breathing room for native memory allocations.

3. **Monitor Native Memory:** Use the Native Memory Tracking (NMT) feature to gain visibility into non-heap memory usage.

```
# Enable NMT in the container  
-XX:NativeMemoryTracking=summary (or detail)
```

Hands-on Activity: Analyzing Off-Heap Requirements

This activity guides you through identifying the native memory footprint of your Java process.

Step 1: Enable NMT and Inspect Memory

Use **jcmd** to inspect the memory distribution of your running containerized application.

1. Access the terminal of your running pod:

```
oc rsh <pod-name>
```

2. Run the NMT summary command:

```
jcmd 1 VM.native_memory summary
```

Example output:

```
sh-5.3$ jcmd 1 VM.native_memory summary
1:
Native memory tracking is not enabled
sh-5.3$
```



If you see "Native memory tracking is not enabled", you must enable it by adding the following JVM option:

```
-XX:NativeMemoryTracking=summary
```

1. Delete the current deployment

```
oc delete deployment my-java-app
```

2. Create a deployment configuration to enable NMT in the container (example below will never work, it is just an example - you need an actual application):

```
cat > my-java-app.yaml << EOF
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-java-app
spec:
  replicas: 1

  selector:
    matchLabels:
      app: my-java-app

  template:
    metadata:
      labels:
        app: my-java-app

    spec:
      containers:
        - name: java-app
          image: quay.io/hummingbird/openjdk:latest

          env:
            - name: JAVA_TOOL_OPTIONS
              value: "-XX:NativeMemoryTracking=summary"
EOF
```

3. Create the deployment:

```
oc apply -f my-java-app.yaml
```

4. Access the terminal of your new running pod:

```
oc rsh <pod-name>
```

5. Inspect the memory distribution:

```
jcmd 1 VM.native_memory summary
```

Example output:

```
sh-5.3$ jcmd 1 VM.native_memory summary
Picked up JAVA_OPTS_APPEND: -XX:NativeMemoryTracking=summary
1:

Native Memory Tracking:

(Omitting categories weighting less than 1KB)

Total: reserved=33545316KB, committed=288676KB
      malloc: 59876KB #80575, peak=111544KB #81451
      mmap:   reserved=33485440KB, committed=228800KB

-          Java Heap (reserved=31424512KB, committed=147456KB)
              (mmap: reserved=31424512KB, committed=147456KB, at peak)

. . .

-      String Deduplication (reserved=1KB, committed=1KB)
              (malloc=1KB tag=String Deduplication #8) (at peak)

-      Object Monitors (reserved=75KB, committed=75KB)
              (malloc=75KB tag=Object Monitors #385) (at peak)
```

1. Analyze the output. Look specifically at:

- **Internal:** Memory used by the JVM internal structures.
- **Symbol:** Memory used for string interning.
- **Thread:** Memory consumed by stack spaces.

Step 2: Simulate Constraint Pressure

To ensure your configuration is resilient, verify that your application does not exceed the container limit.

1. View the current cgroup memory limit for your container:

```
## FIXME: The below path is not found in the container.  
cat /sys/fs/cgroup/memory/memory.limit_in_bytes
```

2. Compare this value against your calculated Heap + Off-Heap usage. If your application's **Total** committed memory (as seen in NMT) + heap usage exceeds this limit, one must adjust your **MaxRAMPercentage** downward or increase the container's memory request/limit in OpenShift **Deployment** YAML.

Troubleshooting Tip

The following risks are possible on the deployment in orchestrators like Openshift 4:

- **OOMKill**: The full memory is above the container size or cgroups v2 detection is failing.
- **OOMException**: Usually off-heap allocation or metaspace, not that if **ExitOnOutOfMemoryError** flag is enabled, so then the JVM will exit.
- **SIGTerm**: In case communication is impacted or the JVM is unresponsive sigterm can kick in. That will be in the events.

If you encounter **cgroup OOMKills** despite having a small Java Heap, use **jcmd** to identify the category consuming unexpected amounts of native memory. From that, investigate off-heap components such as non-container aware libraries, thread pool sizing, and do an investigation from the off-heap size perspective.

Best Practices for Java Memory Configuration in Containers

As Java has evolved to be flexible and container-aware, some practices could be considered anti-patterns, such as:

- **Xmx** direct usage: (in most cases) is an anti-pattern. By default Java will have already percentage utilization inside a container. In other words, the manual strategy of defining absolute heap sizes (e.g., **-Xmx2g**) has become a deployment anti-pattern, because it decouples from the container size.
- JVM arguments directly at build time, rather than runtime. For example, avoid the usage of JVM arguments hardcoded in a Containerfile (Dockerfile), rather than customizable in a deployment resource (YAML), which is more adequate.

When the JVM is unaware of its container boundaries, it will default to host-level metrics, leading to resource contention, poor performance, or premature **cgroups** OOM-Kills.

The current approach focuses on **relative size configuration** so the size of Java's heap changes based on the container size (dynamically requiring restarting). This maintains a healthy balance between the JVM heap and the native (off-heap) memory required by the container environment.

Moving from Absolute to Relative Configuration

To ensure your Java application scales gracefully within an OpenShift or Kubernetes pod, follow these best practices for memory management.

1. Use Percentage-Based Flags

Instead of hardcoding memory values, utilize percentage-based flags. These ensure the JVM dynamically adjusts its heap size relative to the memory limits defined in your Pod specification.

- **-XX:MaxRAMPercentage=N**: Set by default. Defines the maximum heap size as a percentage of the container's memory limit. For most applications, a value between **50.0** and **80.0** is usually adequate, leaving the remaining space for off-heap requirements. Off-heap heavy application may require a low percentage.
- **-XX:InitialRAMPercentage=N**: Sets the initial heap size as a percentage of the container limit.
- **-XX:MinRAMPercentage=N**: Sets the minimum heap size when the container limit is very small.

2. Understanding the Off-Heap allocation

The memory not occupied by the JVM heap is not wasted — it is necessary for the stability of the Java application. When you configure your heap, you must leave sufficient headroom for:

- **Metaspace**: Stores class metadata.
- **Code Cache**: Stores JIT-compiled native code.
- **Thread Stacks**: Each thread requires native memory; a large number of threads can lead to unexpected native memory pressure.
- **Direct Byte Buffers**: Often used by frameworks like Netty, AMQ, or Data Grid.
- **Garbage Collector Metadata**: The structures required by G1GC or ZGC to track object references.

Best Practice: If your application relies heavily on off-heap memory (e.g., streaming or high-throughput messaging), use a lower **MaxRAMPercentage** (e.g., 50%) to prevent the total container usage from hitting the **cgroups** limit.

3. Avoid MaxRAM Override

Avoid setting the **-XX:MaxRAM** flag. When you set **MaxRAM** manually, you also effectively decouple the JVM's view of memory from the actual container **cgroups** limit. This removes completely the purpose of container awareness and can lead to the JVM attempting to allocate more memory than the container is permitted to use, resulting in an immediate OOM-Kill by the kernel.

4. Aligning with Container Orchestration

Ensure your Pod resource requests and limits are configured to support your Java workload:

- **Memory Limits**: Always set a memory limit in your Deployment/DeploymentConfig. The JVM uses this value to calculate the heap size when using percentage-based flags.

- **Avoid Oversizing:** Do not set the limit significantly higher than the application's actual footprint. While Kubernetes handles overcommitment, the JVM will calculate its heap based on the **limit**, not the actual usage.

Summary Table: Configuration Comparison

Approach	Recommendation
<code>-Xmx</code> (Absolute)	Discouraged. Will cause a decoupling and potential OOM-Kills. Also any change will cause problems. <code>Xmx</code> overwrites any percentage (<code>MaxRAMPercentage</code>).
<code>-XX:MaxRAMPercentage</code>	Recommended. Automatically aligns heap with container <code>cgroups</code> limits.
<code>MaxRAM</code>	Avoid. Can create conflicting memory views between the OS and the JVM.



When using Red Hat OpenJDK images, the provided `run-java.sh` initialization script sets a specific `MaxRAMPercentage` percentage, currently at 80%. Previously, the `MaxRAMPercentage` was 50%. This value sets the Max Heap at 80% of the container size. The remainder goes to off-heap. Overriding the `MaxRAMPercentage` with a custom value is sufficient to customize the deployment, considering that 80% heap may not be adequate. If the scripts `run-java.sh` is not used, so the default percentage (`MaxRAMPercentage`) will be 25% (for heap) so 75% for off-heap.

Troubleshooting Resource Constraints

If your application is experiencing `OutOfMemoryException` or being killed by the container orchestrator, analyze the environment with the following diagnostic flow:

1. **Check `VM.info` file:** Use Java Diagnostics (`jcmd`) to print all details of Java concerning containers.
2. **Orchestrator and host details:** Verify the events, deployment yaml status, and host logs (`sos_report`), which will record OOMKill events.
3. **Check `cgroups` usage:** Use `cat /sys/fs/cgroup/memory/memory.stat` inside the container to see if the container is hitting its hard limit. This file is usually used for Prometheus/cadvisor output scrapping.
4. **Analyze Native Memory Tracking (NMT):** Enable `-XX:NativeMemoryTracking=summary` to identify if the memory leak is occurring in the heap or in native memory segments (e.g., excessive thread allocation).
5. **Benchmark:** Always performance-test with a memory profiler before finalizing your percentage values to ensure the off-heap headroom is sufficient for peak load.

OpenShift and Kubernetes Integration

In containerized environments, Java applications rely on integration with OpenShift and Kubernetes to manage resource allocation through cgroups. Understanding how these platforms govern resources is critical for stable deployments:

- **Resource Specifications:** Kubernetes uses `requests` and `limits` defined in deployment YAML to manage compute resources. `Requests` guide the scheduler for node placement, while `limits` enforce hard constraints on CPU and memory usage. Java only cares about limits.
- **Cgroup Enforcement:** Platform resource definitions translate directly into Linux kernel cgroup values. CPU limits define execution quotas, while memory limits act as the boundary for the container's total memory footprint.
- **QoS Classes:** OpenShift employs Quality of Service (QoS) classes to handle resource contention. There are three classes: Guaranteed, Burstable, and Best-effort (in order of eviction). While Guaranteed QoS provides stable resource allocation always, it does not prevent cgroup-level OutOfMemory (OOM) kills or OOMException from happening, just puts the application at the end of the eviction process (after burstable and best-effort).
- **Operational Alignment:** Proper integration requires aligning Java heap configurations with container-level constraints to prevent discrepancies between the JVM's view of memory and the actual limits imposed by the OpenShift platform.

Java only cares about limits, so if the deployment does not have limits, it will use the host as limits. Unbounded deployment is a type of deployment that will use the full Host limits.

Configuring Limits and Requests in Red Hat OpenShift

In the context of Red Hat OpenShift Container Platform (OCP), effectively managing resources is a balancing act between scheduling efficiency and runtime performance. To optimize Java applications, it is critical to distinguish how OCP handles resource definitions and how those definitions translate to the Linux kernel and the Java Virtual Machine (JVM).

The Relationship Between Requests and Limits

OpenShift utilizes Kubernetes-native resource management, defined within the container specification of a Deployment or Pod YAML.

- **Requests:** These are used by the OpenShift scheduler to place a Pod on a node. They represent the minimum amount of CPU and memory guaranteed to the container. If a node does not have enough unallocated resources to satisfy the request, the scheduler will not place the Pod on that node.
- **Limits:** These define the hard ceiling for resource consumption. If a container exceeds its memory limit, it risks being terminated by the kernel (OOMKilled). CPU limits act as a throttling mechanism, restricting the container's execution time within a defined period.

Resource Allocation Table

Resource Type	Purpose	Calculation Method
CPU Request	Scheduling and node assignment.	Sets <code>cpu.share</code> in cgroups to prioritize usage.
CPU Limit	Binding execution allocation (throttling).	Sets quota and period (defaults to 100ms segments).

Perspectives on Resource Awareness

It is vital to understand that different layers of your infrastructure interpret these settings differently:

1. **OpenShift/Kubernetes Layer:** The scheduler uses **requests** to manage cluster density and node capacity. It ignores **limits** for scheduling decisions.
2. **Kernel Layer:** The kernel enforces **limits** via Control Groups (cgroups). Quotas and periods derived from the container spec are translated into filesystem values in `/sys/fs/cgroup`.
3. **Java Layer (OpenJDK):** Modern versions of OpenJDK (post-8u191) are container-aware. However, **Java only considers limits, not requests**. Even if you define CPU requests in your YAML, the JVM will ignore them for internal heap sizing and thread pool calculations.



While Java workloads are burstable (threads dynamically demand kernel time), the JVM performs its cgroup reading at **startup** only once. Consequently, the JVM is inelastic, i.e. changing a container limits dynamically and expect the JVM to change its settings without restart. If the user changes limits at runtime without restarting the Pod, the JVM will **not** automatically adjust its internal heuristics to match the new constraints.

Best Practices for OCP Configurations

- **Avoid Overcommitting Critical Workloads:** For production-critical applications, consider Guaranteed Quality of Service (QoS) - this will put the application as the last to be evicted. A pod is considered Guaranteed when both requests and limits are set to the same value for all containers in the pod.
- **Avoid Undercommitting Workloads:** Benchmark the application to use the correct heap percentage (MaxRAMPercentage) to make sure heap and off-heap have space to grow accordingly.
- **Limit vs. Request Parity:** If you are unsure of the application's baseline, start by setting requests lower than limits to allow for burstable performance, but ensure the limit is high enough to accommodate peak load to prevent CPU throttling.
- **Memory Awareness:** Prefer using memory limits into containers to prevent a single Java container from consuming a large chunk of available node memory and CPU, which can impact other pods on the same node. Unbounded deployments can be very useful, but need to be used in specific scenarios.

Unbounded deployment

Although not the core of this course, it is totally possible and sometimes even desired to have Java applications without container limits, the so called unbounded. Scenarios such as very few java applications, large OCP nodes, and specific scenarios for high spike demand can justify its utilization.

This is achieved by creating an deployment without limits, which will have two main consequences:

- * Java will use the full host as limit: which can make the application use a very large number of CPUs as baseline (for thread pools) and a large memory utilization (given heap will be based on the host value).
- * QoS will be Burstable or Best Effort, depending if requests are set.

This type of application deployment should be treated with caution, given the fact that several applications (unbounded) will cause a race for the hosts resources and may impact directly on CPU throttling (many threads competing at the same time for CPU and memory resources). Consequently, this can overload the host and create more possibilities for eviction scenarios (several Java unbounded demanding more and more resources) causing generalized evictions.

Hands-on Activity: Inspecting cgroup Resource Constraints

In this activity, you will verify how OpenShift translates your YAML configuration into cgroup settings visible to the Java application.

Prerequisites:

- An active OpenShift cluster (OCP 4 for Openshift Operators deployments).
- **oc** CLI authenticated with administrative or project-level access.

Note: When deploying with an Operator, the Operator will control the deployment directly, therefore the user needs to modify the settings via the respective Custom Resource.

Steps:

1. **Deploy a sample Java application** with defined limits and requests (below is just an example - it won't deploy)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-resource-demo
spec:

  selector:
    matchLabels:
      app: java-resource-demo

  template:
    metadata:
      labels:
        app: java-resource-demo
    spec:
```

```
containers:
- name: java-resource-demo
  image: quay.io/hummingbird/openjdk:latest
  resources:
    requests:
      memory: "256Mi"
      cpu: "500m"
    limits:
      memory: "512Mi"
      cpu: "1000m"
```

2. **Access the running Pod** to inspect the cgroup limits:

```
oc rsh <pod-name>
```

3. **Check the CPU quota** enforced by the kernel:

```
# FIXME: The below file is not found in the container
$ cat /sys/fs/cgroup/cpu/cpu.cfs_quota_us
# This will return 100000 (100ms), confirming the 1000m CPU limit.
$ jcmd 1 VM.info | grep active_processor_count
```

Example Output:

```
active_processor_count: 1 <-- shows 1 CPU
```

4. **Observe Java's internal detection** of these limits:

```
java -XshowSettings:vm -version
```

Example output:

```
VM settings:
  Max. Heap Size (Estimated): 123.75M
  Using VM: OpenJDK 64-Bit Server VM

openjdk version "25.0.4.1" 2026-08-18
OpenJDK Runtime Environment (Red_Hat-25.0.4.1.1-0) (build 25.0.4.1+1)
OpenJDK 64-Bit Server VM (Red_Hat-25.0.4.1.1-0) (build 25.0.4.1+1, mixed mode,
sharing)
```

FIXME: Limit is set to 512Mi but the heap size displayed is 123.75M. Output contradicts the below explanation.

Look for the output related to **Max. Heap Size** and **active_processor_count**. You will notice these values are derived from the limits, not the requests.

Understanding cgroups implications for Java applications

In the context of modern container orchestration, the Linux kernel's **control groups (cgroups)** feature acts as the fundamental mechanism for resource isolation. For Java applications running in OpenShift or Kubernetes, cgroups provide the enforcement boundaries for CPU and memory usage.

Understanding how the Java Virtual Machine (JVM) interacts with these cgroups is critical for preventing performance degradation and application crashes.

The Role of cgroups in Resource Management

Cgroups allow the kernel to organize processes into hierarchical groups and allocate resources—such as CPU time, system memory, and network bandwidth—to these groups. When you define **limits** and **requests** in your OpenShift Deployment configuration, OpenShift translates these values into cgroups filesystem entries (typically found under `/sys/fs/cgroup/`). Details:

- cgroups v1: `/sys/fs/cgroup/memory/memory.max_usage_in_bytes`
- cgroups v2: `/sys/fs/cgroup/memory.max`

CPU and Memory Enforcement

- **CPU Limits:** Java applications are inelastic regarding CPU detection - it won't detect every 10s, it will detect once at start-up. Upon startup, the JVM queries the cgroup filesystem to determine how many CPU cores are available. It uses this information to size its internal thread pools (e.g., the Common ForkJoinPool). Once started, this value generally remains fixed. If a container is restricted via a quota, the kernel throttles the CPU cycles available to the JVM.
- **Memory Limits:** The JVM monitors the `memory.limit_in_bytes` (cgroups v1) or `memory.max` (cgroups v2) files. If the JVM is container-aware, it uses these values to calculate default heap sizing. The exact percentage will depend on the version and method of starting: 25% for deployment without `run-java.sh`, 80% for `run-java.sh` script (current Red Hat OpenJDK images) and 50% for `run-java.sh` script started (old Red Hat OpenJDK images).

The Danger of Decoupling

The most significant risk for Java applications in containers is the **decoupling of memory settings** from container limits. This occurs when the JVM is either non-container-aware (older versions) or is manually configured with fixed values that ignore the container's environment.

When the JVM heap size (`-Xmx`) is set higher than the actual memory limit enforced by the cgroup:

1. **The Invisible Limit:** The JVM believes it has more memory available than the container actually provides.
2. **Kernel Intervention:** When the application attempts to allocate memory beyond the cgroup

limit, it does not trigger a standard `java.lang.OutOfMemoryError`. Instead, the **cgroup OOM Killer** invokes the kernel to terminate the process immediately.

3. **Observability Gaps:** Because the kernel performs the termination, these events often do not appear as standard Java application exceptions or OpenShift deployment events. This leads to silent failures, node-level instability, or a crash-looping pod that provides little diagnostic information.

Best Practices for cgroup-Safe Java

To ensure your Java applications coexist healthily with cgroup limits, adopt the following strategies:

- **Utilize Container-Aware JVMs:** Always use Java 8u191+ or modern versions (Java 11, 17, 21+), which include the JDK8146115 enhancement. These versions automatically read the cgroup filesystem to adjust memory ergonomics.
- **Use Relative Sizing:** Instead of hardcoding `-Xmx` values (e.g., `-Xmx2g`), use container-aware flags like `-XX:MaxRAMPercentage`. This allows the JVM to scale its heap dynamically based on the memory limit assigned to the container (e.g., `-XX:MaxRAMPercentage=75.0` will set the heap to 75% of the container limit).
- **Account for Off-Heap Memory:** Remember that the total memory used by a container includes the heap, the Metaspace, stack space for threads, code cache, and native buffers (Direct Byte Buffers). Ensure the percentage (`MaxRAMPercentage`) leaves sufficient room for heap to grow and for non-heap segments, or the container will exceed its cgroup limit triggering an OOMKill.



When debugging, always check the `memory.stat` file inside the container (if permissions allow) or use tools like `jcmd` (`VM.info`) to inspect current memory usage and compare them against the deployment's defined resource limits.

Navigating Kubernetes resource management for Java

In Kubernetes and OpenShift environments, resource management is governed by the orchestration layer: kubeAPI, which creates deployments with CPU settings. In case this fails (at container creation or deployment) events will show the cause. * The specific cgroups boundaries are maintained by the Host kernel. In case this fail (memory grows too much) the kernel will trigger an OOMKill event. * For Java applications, the primary challenge lies in the abstraction gap: the JVM manages its own memory (heap, code cache, metaspace) and CPU (GC threads, JIT compiler threads). In case this fails, the JVM will cause `OOExceptions` or Errors (crashes). * Kubernetes manages the container's total resource footprint and keeps the overall applications running nominally via probes, which in case they fail cause `SIGTERMs`.

Resource Requests vs. Limits

Effective resource management requires a clear understanding of the difference between requests and limits:

- **Requests:** The guaranteed amount of resources (CPU/Memory) allocated to the container. The Kubernetes scheduler uses these values to place pods on nodes.

- **Limits:** The maximum threshold allowed for the container. If a Java application exceeds its memory limit, the Linux kernel OOM (Out of Memory) Killer will terminate the process.



Setting **Limits** equal to **Requests** results in a **Guaranteed Quality of Service (QoS)** class. While this provides the highest priority, it does not prevent the OOM Killer from terminating the pod if the JVM heap plus off-heap memory usage exceeds the container limit.

Cgroups and the Java JVM

Since the introduction of JDK 8u191 (and the backport of **JDK8146115**), the JVM has become container-aware. It reads cgroup filesystem metrics to determine available resources rather than querying the host OS kernel.

When the JVM starts, it calculates heap ergonomics based on the container's cgroup limits:

- **Memory:** If no **-Xmx** is explicitly set, the JVM defaults to a fraction of the container's memory limit - see details on the percentage below.
- **CPU:** The JVM detects the number of available CPUs via cgroup shares/quotas and scales the number of Garbage Collection (GC) threads accordingly.

The Impact of Dynamic Resizing

While recent Kubernetes versions (1.27+) and OpenShift 4.4+ introduced features to resize pod resources without a restart, the JVM behavior remains tied to its initialization state.

1. **Startup Snapshot:** The JVM performs its initial heap and thread pool calculations based on the resource values present when the process is invoked.
2. **Detection vs. Adaptation:** While the JVM can technically observe changes in the cgroup filesystem during runtime, it does not resize its internal heap boundaries (Xmx) dynamically.
3. **Risk:** If a container is vertically scaled (e.g., memory limit increased), the JVM will likely continue to respect the original heap size calculated at boot, potentially leaving the application under-provisioned despite the increase in available container resources.

Best Practices for Java on Kubernetes

To ensure stability in a Kubernetes-managed environment, adhere to the following configurations:

- **Explicit Heap Definition:** Even with container awareness, Java is already container aware and this ensures the heap scales proportionally if the container limit is adjusted in the deployment manifest. The best practice is for the user to tune this value (e.g., **-XX:MaxRAMPercentage=75.0**), rather than setting directly Xmx, which overwrites anything else.
- **Account for Off-Heap Memory:** Remember that the total container memory limit must accommodate the Java heap, Metaspace, Code Cache, Thread stacks, and Native memory (NMT). Tune **-XX:MaxRAMPercentage** conservatively (e.g., 60-75%) to prevent OOM Exceptions events at heap level.
- **CPU Throttling:** Be aware that setting strict CPU limits can lead to CPU throttling, where the

kernel pauses threads when the quota is reached. Monitor `container_cpu_cfs_throttled_periods_total` in Prometheus to identify performance bottlenecks. The application's threads will be paused during this throttling process, which will show as a plateau in Prometheus CPU usage.



Always use the latest stable version of Red Hat's OpenJDK, as it includes the most robust cgroup v2 support and the latest optimizations for container environment interactions.

Troubleshooting and Tooling

Troubleshooting Java in containerized environments requires distinguishing between tools that report host-level node metrics versus those that are container-aware. Improper resource utilization undoubtedly will lead to memory misconfiguration, `OOMKills`, `OOMExceptions`, `SIGMTERMs`, and under/over-utilization of resources.

- **Host-level vs. Container-aware Tools:** Standard Linux utilities like `lscpu`, `nproc`, and `free -m` reflect the underlying node's resources rather than container limits.
- **Container-aware Diagnostics:** Utilize JVM-specific tools such as `jcmt`'s `VM.info` or `-XX:+PrintFlagsFinal` (via `--XshowSettings:system`) to accurately retrieve the memory and CPU constraints enforced on the container.
- **Deployment Inspection:** Analyze pod logs to verify if scripts like `run-java.sh` are active or if the default OpenJDK memory allocation is scaling correctly.
- **Heap Analysis:** Compare the JVM's reported heap size against container limit values to identify potential discrepancies between configured `-Xmx` values and actual resource availability.
- **Resource Tuning:** Be aware that manually overriding `-Xmx` on container-aware JVMs can cause mismatches with container limits, potentially leading to container underutilization or runtime `OOMExceptions`.

Identifying Container-Aware versus Non-Container-Aware Tools

When managing Java applications in Red Hat OpenShift or Kubernetes, understanding whether your diagnostic tools perceive the underlying host node or the isolated container environment is critical. Misinterpreting these outputs often leads to OutOfMemory (OOM) kills or CPU throttling because the application may be relying on node-level visibility rather than container-level boundaries.

Distinguishing Tool Behavior

To effectively troubleshoot Java in containers, you must categorize your diagnostic tools based on their ability to parse `cgroups` (control groups).

Non-Container-Aware Tools

Non-container-aware tools report information regarding the physical or virtual host (the OCP Node). Relying on these tools within a pod will provide a false sense of resource availability.

- `lscpu`: Reports the total CPU architecture and count of the physical worker node, not the specific CPU quota assigned to your pod.
- `nproc`: Returns the number of processors available to the host OS, ignoring the Kubernetes `resources.limits.cpu` defined in your pod specification.
- `free -m`: Displays the total RAM available to the node. If your pod has a 512MB limit but the node has 64GB, this command will inaccurately suggest you have 64GB available for your heap.

Container-Aware Tools

Container-aware tools interface with the Linux `cgroups` subsystem to identify the specific constraints imposed by the orchestrator. These are the "source of truth" for your application environment.

- `-XX:+PrintFlagsFinal` / `-XshowSettings:system`: These flags instruct the JVM to report the memory and CPU settings it has calculated based on container limits. Use these to verify if the JVM correctly identified the container memory boundary.
- `VM.info` (via `jcmd`): Provides a snapshot of the JVM, including flags and internal settings that reflect whether the container-aware logic was successfully applied.
- `run-java.sh`: Always present in Red Hat OpenJDK images, sets certain specific Java flags, such as `ParallelGC` and `MaxRAMPercentage` Java heap sizes automatically before the JVM starts. In current images, this script won't make Java container aware (that is on the Java code itself) but it will make a difference on the Heap percentage that comes by default (`MaxRAMPercentage`) value, in case it is not used the default OpenJDK container should set heap at 25% of the deployment (upstream default value).

Diagnostic Strategy

To verify if your environment is correctly configured, compare the output of host-level tools against JVM-internal tools.

Tool	Scope	Usage Case
<code>free -m</code>	Node	Use only to verify the underlying node capacity.
<code>nproc</code>	Node	Ignore when calculating thread pools or garbage collection threads.
<code>java -XshowSettings:system</code>	Container	Use to confirm the JVM sees the pod's <code>limits.memory</code> .
<code>jcmd <pid> VM.info</code>	Container	Use for deep runtime inspection of detected resource limits.

Detecting Misconfiguration

If your application is crashing with an `OOMKiller` signal, follow this diagnostic workflow:

1. **Check JVM visibility:** Run `java -XshowSettings:system` inside the running container.
2. **Compare limits:** Look at `MaxHeapSize` and `TotalPhysicalMemory` in the output.
3. **Evaluate OCP Manifests:** Compare these values against your OpenShift `Deployment` or `DeploymentConfig` resource requests and limits.
4. **Log Analysis:** Review the pod startup logs. If the logs show that the JVM is using a default (e.g., 25% of physical memory) rather than a percentage of the container limit, the JVM is likely not properly detecting the `cgroup` constraints.



If the application logs indicate it is defaulting to 25% of the **node's** memory rather than the **container's** limit: verify the OpenJDK version (8u191+) is correctly detecting containers limits, e.g. for cgroups v2 compliance. If the application logs indicate it is defaulting to 25% of the **container's** memory rather than 50% or 80%: verify if the application is deployed with `run-java.sh` as ENTRYPOINT (that's the default).

Analyzing Container Limits and Resource Utilization

To ensure Java applications perform predictably within Red Hat OpenShift or Kubernetes, you must understand how the JVM interacts with the underlying control groups (cgroups). When an application is container-aware, the JVM can query the cgroup filesystem to determine its actual boundaries rather than defaulting to the physical host's resources.

Understanding Resource Visibility

The primary challenge in containerized Java is the discrepancy between what the Operating System sees (physical host) and what the container runtime enforces (cgroup limits).

The Mechanics of Detection

Container-aware JVMs (starting from Java 8u191 and newer) automatically inspect cgroup files, such as `memory.limit_in_bytes` to size the heap, but cgroups will also be used to determine the available processor count (for thread pool calculations).

When analyzing your application, you can extract this metadata using `jcmd`. If your JVM is properly detecting the container environment, the output will reveal specific cgroup configurations:

```
# Example output from jcmd <pid> VM.check_commercial_features or custom diagnostic
commands
container_type: cgroupv1
active_processor_count: 1
cpu_quota: -1
memory_limit_in_bytes: 536870912
memory_usage_in_bytes: 125108224
```

- **active_processor_count**: Essential for the JVM to size its internal thread pools (e.g., ForkJoinPool).
- **memory_limit_in_bytes**: Dictates the physical memory limit the JVM uses to calculate its initial heap size (typically 1/4 of the container limit by default).

Identifying Limitations and Misconfigurations

Analyzing resource utilization requires verifying that your JVM heap plus off-heap requirements remain within the Kubernetes/OpenShift **limits**.

The Decoupling Risk

There are several risks of decoupling. - The Xmx set the full size of the heap, given it is unrelated so the container size can reduce and starve the application or have too much space useless. - A common pitfall is ignoring the off-heap footprint. If the Java Heap (Xmx) to 80% of your container limit, but your application requires 30% for off-heap memory (Metaspace, Code Cache, Thread stacks, Direct Buffers), the container will exceed its memory limit and OOMKill.

This results in a **cgroup OOMKill**, which is distinct from a JVM `java.lang.OutOfMemoryError`.

Troubleshooting Checklist

When analyzing resource constraints, follow these steps:

1. **Compare Requests vs. Limits:** Ensure your OpenShift `requests` are sufficient for startup, while `limits` accommodate peak load.
2. **Verify Cgroup Awareness:** Use `jcmd` to verify if the JVM identifies the correct `memory_limit_in_bytes`. If this shows `-1` or reflects host memory, the JVM is not container-aware, and you must explicitly configure `-XX:MaxRAMPercentage`.
3. **Monitor via Tools:**
 - **Container-aware:** Use `kubectl top pod` or the OpenShift Web Console metrics.
 - **Non-container-aware:** Avoid using legacy `top` or `free` commands inside the container, as they often report host-level data rather than container-constrained data.

Hands-on Activity: Inspecting JVM Resource Awareness

FIXME: This can be general instructions and not hands-on activity to be followed by learner.

In this activity, you will verify if your currently running pod is correctly interpreting its cgroup limits.

1. **Access the running pod:**

```
oc rsh <pod-name>
```

2. **Identify the PID of the Java process:**

```
jps (or ps -aux, or jcmd)
# Output: 1 AppMain
```

3. **Run the diagnostic command to check resource visibility:**

```
jcmd 1234 VM.native_memory summary
# Look for container-specific configuration in the JVM diagnostic output
```

4. **Cross-reference with OpenShift limits:** Compare the `memory_limit_in_bytes` reported by the JVM with the memory limit defined in the deployment YAML. If they do not align, verify if the application is container aware, it is using `java-run.sh`. In case needed, one can adjust the `JAVA_OPTS`` and `JAVA_OPTS_APPEND` using `-XX:MaxRAMPercentage` in case `java-run.sh` is used. The environment variables: `JAVA_OPTS` and `JAVA_OPTS_APPEND` can be used as long as `java-run.sh` are used.

```
# Example:
$ podman run --memory-reservation=700m --memory=500m --memory-swap=1000m --rm -it
localhost/openjdk:app
memory_limit_in_bytes: 512000 k <----- memory 500mb
memory_and_swap_limit_in_bytes: 1024000 k <----- swap and memory 1000mb
memory_soft_limit_in_bytes: 716800 k <----- soft limit 700mb
```



If you observe `cgroup OOMKills` despite the JVM reporting correct memory usage, analyze your off-heap usage via `jcmd <pid> NativeMemoryTracking=summary`. You may need to decrease your `MaxRAMPercentage` to provide more "breathing room" for the native memory footprint.

Diagnostic strategies for Java deployments in OpenShift

When managing Java applications in OpenShift, traditional diagnostic tools often fail because they lack container-awareness capabilities. Tools that look at the host operating system's total resources rather than the container's constrained cgroup limits can lead to misleading metrics. Effective diagnostics require tools and strategies that respect the isolation boundaries defined by Kubernetes.

Identifying Container-Aware Tools

A critical diagnostic step is distinguishing between container-aware and legacy tools.

- **Legacy Tools:** Tools like `top`, `free`, or older JVM monitoring utilities may report the host's total RAM and CPU count. Relying on these leads to "false positives" where an administrator believes the application has more resources than it actually does.
- **Container-Aware Tools:** Modern JDKs (8u191+, 11+) and utilities like `jcmd`, `jstat`, and modern observability agents are cgroup-aware. They query the container's `/sys/fs/cgroup/` filesystem to report limits specifically applied to the pod.

Diagnostic Strategies

To effectively diagnose resource constraints or performance issues, follow these strategies:

1. Validating Resource Visibility

Always verify that the JVM sees the limits defined in your OpenShift `DeploymentConfig` or `Deployment`.

You can inspect this by executing into the pod and checking the container's perceived memory and CPU:

```
# Check perceived max heap from within the pod
oc exec <pod-name> -- java -XshowSettings:vm -version
```

Check if `MaxHeapSize` is calculated correctly based on the container memory limit rather than the worker node's total memory.

2. Utilizing `jcmd`

`jcmd` is the gold standard for Java diagnostics within OpenShift. It allows you to perform heap dumps, thread dumps, and analyze VM flags without requiring a restart.

- **Thread Dumps:** Identify if the application is hitting thread limits or experiencing deadlocks.
- **GC Performance:** Use `jcmd <pid> GC.run` to trigger garbage collection or `jcmd <pid> VM.flags` to verify if `-XX:MaxRAMPercentage` is correctly configured.

3. Analyzing cgroup Constraints

If a pod is frequently restarting, it may be hitting OOM (Out of Memory) kills. Since the OOM Killer is enforced at the cgroup level, standard Java `OutOfMemoryError` logs might not appear if the kernel terminates the process before the JVM can report it.

- **Inspect Pod Events:** Use `oc describe pod <pod-name>` to check for the `OOMKilled` state.
- **Check Exit Codes:** An exit code of `137` is a strong indicator that the container was terminated by the OOM killer because it exceeded the OpenShift memory limit.

Hands-on Lab: Analyzing Resource Constraints

In this lab, we will use `jcmd` to inspect a running Java process and verify its awareness of container limits.

Prerequisites

- An active OpenShift cluster with a running Java pod.
- `oc` CLI configured with cluster access.

Step 1: Verify JVM resource settings

Execute into the pod to check how the JVM views the memory limits.

```
# Run the VM settings check
oc exec <pod-name> -- java -XX:+PrintFlagsFinal -version | grep MaxHeapSize
```

Observe the output. If you have defined a `memory` limit of 512Mi in your manifest, the `MaxHeapSize` should be a fraction of that limit (if using `MaxRAMPercentage`).

Step 2: Use `jcmd` to analyze the process

Access the container shell to utilize `jcmd`.

```
oc rsh <pod-name>

# List running Java processes
jcmd

# Get a detailed report of the VM's perceived memory constraints
jcmd <pid> VM.flags
```

Step 3: Troubleshooting a constraint

If the application is slow, check if CPU throttling is occurring due to the Kubernetes `cpu` limit:

1. Check the `cpu.stat` file within the container:

```
[source,bash]
----
cat /sys/fs/cgroup/cpu/cpu.stat
----
```

2. Look for `nr_throttled`. If this number is increasing rapidly, your CPU `limit` is too low, and the kernel is throttling your application, causing latency.

Best Practices Summary

- **Never rely on `free` or `top`** inside a container for memory capacity planning.
- **Always use `-XX:MaxRAMPercentage`** instead of fixed `-Xmx` values to ensure the heap scales dynamically with the container limit.
- **Prioritize `jcmd`** for all runtime analysis, as it is native to the OpenJDK and fully supports container-cgroup awareness.

Java Diagnostics: `jcmd`

The `jcmd` utility is a versatile diagnostic tool included with the JDK. In containerized environments, it is indispensable for inspecting the internal state of a running Java Virtual Machine (JVM) without requiring external profilers or complex agent attachments.

Role of Java Diagnostics (`jcmd`) in Containerized Environments

In a containerized world, `jcmd` allows developers and operators to bridge the gap between the container's resource limits (cgroups) and the JVM's internal resource allocation. When troubleshooting performance or memory issues in OpenShift or Kubernetes, `jcmd` provides a direct view into the JVM's view of the container.

Key Capabilities

- **Process Identification:** List running Java processes within the container.
- **Performance Diagnostics:** Retrieve thread dumps, GC statistics, and class loading metadata.
- **Memory Management Analysis:** Inspect heap usage and native memory tracking (NMT).
- **Dynamic Configuration:** Adjust certain JVM flags at runtime (where supported).

Using jcmd for Resource Inspection

The main tool for Java investigations inside a container is `VM.info`, which is available in JDK 11+. It prints several details on the Java process itself, from the JVM perspective. See full review on [Interpreting VM.info file in OpenJDK/OracleJDK](#).

When a container is suffering from `OutOfMemory` (OOM) errors or unexpected latency, you can execute `jcmd` directly inside the pod to gather data.

```
# List all running Java processes
jcmd

# Java full dump (JDK 11+)
jcmd <PID> VM.info

# Retrieve a thread dump for a specific PID
jcmd <PID> Thread.print

# Check GC performance statistics
jcmd <PID> GC.run
```

Analyzing Native Memory

One of the most critical aspects of containerized Java is distinguishing between the **Java Heap** and **Off-Heap memory**. If the container is being killed by the OOMKiller but the heap appears healthy, the issue often resides in native memory (e.g., Metaspace, code cache, or direct buffers).

Enable Native Memory Tracking (NMT) via the JVM argument `-XX:NativeMemoryTracking=summary` to use `jcmd` to identify leaks:

```
# View native memory summary
jcmd <PID> VM.native_memory summary
```

Best Practices for Diagnostics in OpenShift

1. **Accessing the Pod:** Use `oc rsh <pod-name>` to gain shell access to the running container.
2. **Tooling Availability:** Ensure your container base image includes the `jcmd` binary (standard in most OpenJDK-based images).
3. **Non-intrusive Data Collection:** Use `jcmd` for point-in-time snapshots. Avoid high-frequency

polling, as diagnostics can consume CPU cycles within the container's limited quota.

4. **Verification:** Compare the output of `GC.heap_info` against your pod's `resources.limits.memory` to verify if the JVM is respecting the container's memory boundaries.

Hands-on Lab: Inspecting JVM Resources

FIXME: Can this also be general instructions instead of hands-on activity?

In this exercise, we will verify the JVM's memory awareness.

1. **Access the container terminal:**

```
oc rsh deployment/my-java-app
```

2. **Identify the PID of the Java process:**

```
jcmd -l
```

3. **Inspect the current heap configuration:** Use the following command to check how the JVM is interpreting the container memory limits:

```
jcmd <PID> VM.flags
```

Look for `-XX:MaxHeapSize` or `-XX:InitialHeapSize`.

4. **Analyze GC performance:** Check if the Garbage Collector is struggling due to container constraints:

```
jcmd <PID> GC.class_histogram
```

Troubleshooting Tip: If `jcmd` reports a heap size significantly different from your OpenShift memory limits, check your deployment YAML to ensure you are not manually overriding the heap size with `-Xmx`, which might be conflicting with the container-aware auto-tuning.

Hands-on Lab

The hands-on lab focuses on practical application of container awareness principles within an OpenShift environment. Key activities include:

- **Resource Tuning:** Adjusting Java heap settings using percentage-based configurations to align with container-imposed limits.
- **Monitoring:** Utilizing container-aware diagnostic tools to track actual CPU and memory utilization rather than host-level metrics.
- **Troubleshooting:** Applying diagnostic strategies—such as inspecting pod configurations and resource constraints—to resolve deployment issues within OpenShift.

Tuning Java Heap Settings Based on Container Limits

In containerized environments orchestrators, Red Hat OpenShift, Red Hat OpenJDK is able to detect the container boundaries. Instead, it respects the **cgroup** limits assigned to the container. Tuning the Java heap using percentages is the recommended and standard approach to ensure the JVM remains flexible and associated with the container's resource allocation.

The Shift to Percentage-Based Memory Management

Historically since Java inception, developers used the **-Xmx** flag to set an absolute heap size (e.g., **-Xmx2g**). In a container, this creates a static configuration that ignores the actual memory limit of the container. If the container limit is lowered, the JVM may attempt to exceed it, leading to a **cgroup** OOM-Kill.

By using **RAM Percentage flags**, you decouple the specific memory limit from the application code, allowing the JVM to calculate its heap size dynamically based on the container's available memory.

Key Configuration Flags

- **-XX:MaxRAMPercentage=N**: Sets the maximum heap size as a percentage of the container's memory limit. The default upstream limit is 25%, though specialized container images (like Red Hat's **run-java.sh** scripts) will default to 80%.
- **-XX:InitialRAMPercentage=N**: Sets the initial heap size as a percentage of the container limit.
- **-XX:MinRAMPercentage=N**: Defines the minimum heap size used when the container memory limit is small.



When tuning these percentages, remember that the **total container memory** must accommodate more than just the heap. You must account for **off-heap memory**, which includes:

- Metaspace (Class metadata)
- Thread stacks
- Code Cache

- GC structures and internal JVM native memory
- Off-heap buffers (used by Netty, AMQ, or Data Grid)

Setting `-XX:MaxRAMPercentage` too high (e.g., 95%) leaves insufficient room for native memory, inevitably leading to native memory pressure and `cgroups` OOM-Kill.

Hands-on Lab: Tuning Heap for a Containerized Application

In this lab, you will configure a Java application to use container-aware memory settings instead of hardcoded heap limits.

Prerequisites

- Access to an OpenShift cluster or a local Docker environment.
- A containerized Java application image.

Step 1: Inspect Current Configuration

First, verify that your application is currently running without explicit `-Xmx` flags or is using percentage-based flags. Check the environment variables or the deployment configuration:

```
# Check the deployment environment
oc set env deployment/<deployment-name> --list
```

Step 2: Configure Percentage-Based Heap

Update your deployment configuration to utilize `MaxRAMPercentage`. This ensures that if you increase the pod's memory limit in the future, the heap scales automatically without requiring a code or environment variable change.

Apply the following environment variables to your deployment:

```
oc set env deployment/<deployment-name> \
  JAVA_TOOLS_OPTIONS="-XX:MaxRAMPercentage=75.0 -XX:InitialRAMPercentage=50.0"
```

Step 3: Validate the Configuration

After the deployment restarts, verify that the JVM has correctly calculated the heap size relative to your pod's memory limit.

1. Get the pod name:

```
oc get pods
```

2. Run `jcmd` inside the container to inspect the VM arguments:

```
oc exec <pod-name> -- jcmd 1 VM.flags
```

Observe the output: You should see the `-XX:MaxRAMPercentage` flag active. If you check `VM.max_heap_size`, you will see an absolute byte value calculated based on the 75% ratio of your current container limit. `jcmd VM.info` will bring much more details and container size/settings.

Step 4: Monitoring

Use the OpenShift web console or CLI to monitor memory usage. Compare the Memory Usage metric against the container Limit. Ensure that the usage remains below the total limit, accounting for the gap required for native off-heap memory.



If you observe `java.lang.OutOfMemoryError: Metaspace` or native crashes, your `MaxRAMPercentage` is too aggressive. Lower the value by 5-10% and redeploy.

Reference Documentation

For complete details and subscriber-exclusive guidance on analyzing these diagnostic artifacts, refer to the official Red Hat knowledgebase article:

- [Red Hat Solution 6973111: Interpreting VM.info file in OpenJDK/OracleJDK](#)

Common Properties

The `VM.info` file typically includes: - JVM build details and version strings - Operating system environment specifications - Ergonomics and default sizing parameters

Monitoring CPU and Memory Usage with Container-Aware Tools

When deploying Java applications in containerized environments like Red Hat OpenShift or Kubernetes, traditional Linux diagnostic tools often fall short. Tools that are not "container-aware" report the physical or virtual resources of the entire host node rather than the specific constraints applied to your pod.

This section explores how to distinguish between container-aware and node-aware outputs and how to use them effectively for troubleshooting.

The Visibility Gap

If you use standard Linux utilities such as `lscpu`, `nproc`, or `free -m` inside a container, you are likely seeing the resources of the underlying host node. Relying on these values for capacity planning and testing likely will lead to `OOMExceptions` (`OOMException`), Out of Memory (OOM) kills or CPU throttling because the JVM may attempt to allocate resources exceeding the container's cgroup

limits.

To monitor your Java application accurately, you must utilize tools and flags that interface directly with the **cgroups** filesystem.

Container-Aware Diagnostic Tools

Usage of Java Diagnostics (jcmd) is always preferred.

1. JVM Flags:

The JVM provides built-in mechanisms to report exactly what it sees regarding container limits.

- **--XshowSettings:system**: This is the most critical flag for initial diagnosis. It displays the system information the JVM has detected, including cgroup-enforced memory limits and the active processor count.
- **VM.info**: When using **jcmd**, this command provides a comprehensive dump of the current JVM configuration, including container-specific constraints.

2. Interpreting Container-Aware Output

When running a container-aware command (such as **jcmd**'s **VM.info**'s and **java -XshowSettings:system -version**), look for the following key indicators:

- **active_processor_count**: Represents the number of CPUs available to the JVM based on the Kubernetes/OpenShift CPU limits.
- **memory_limit_in_bytes**: Reflects the specific cgroup memory limit. If this value does not match your expected Kubernetes resource limit, the JVM is not correctly detecting the container boundary.
- **active processor`**: this shows how many CPU were detected.

Example:

```
container (cgroup) information:
container_type: cgroupv2
cpu_cpuset_cpus: 0-3
cpu_memory_nodes: 0
active_processor_count: 3 <----- process count
cpu_quota: 220000 <----- 2.2 cpus
cpu_period: 100000
cpu_shares: no shares
memory_limit_in_bytes: 12582912 k <----- 12gb limit
memory_and_swap_limit_in_bytes: 12582912 k
memory_soft_limit_in_bytes: unlimited
memory_usage_in_bytes: 758948 k
memory_max_usage_in_bytes: 759060 k
kernel_memory_usage_in_bytes: 10624 k
kernel_memory_max_usage_in_bytes: unlimited
kernel_memory_limit_in_bytes: 10628 k
```

```
maximum number of tasks: 98938
current number of tasks: 98
```

Hands-on Activity: Verifying Container Limits

In this activity, you will compare "Node-aware" vs "Container-aware" data to ensure your JVM is correctly bounded.



Ensure you have a running Java pod in your OpenShift/Kubernetes environment and access to `oc rsh` or `kubectl exec`.

Step 1: Compare Node vs. Container Visibility

1. Log into your running Java pod:

```
oc rsh <pod-name>
```

2. Run a non-container-aware command to see the node's resources:

```
free -m
nproc
```

Observe: The output reflects the total RAM and CPUs of the entire OpenShift worker node.

3. Run the container-aware JVM setting command:

```
java -XshowSettings:system -version
```

Step 2: Analyze the JVM Output

Locate the container information section in the output. You should see entries similar to:

```
container (cgroup) information:
  container_type: cgroupv2
  cpu_cpuset_cpus: 0-1
  active_processor_count: 2
  memory_limit_in_bytes: 536870912
```

- **Verification:** Confirm that `memory_limit_in_bytes` aligns with the memory `limit` defined in your pod's YAML configuration.
- **Action:** If these numbers differ from your configured limits, check if your JVM version is 8u191 or later, as older versions lack proper cgroup awareness.

Step 3: Real-time Analysis with jcmd

1. While the application is running, use `jcmd` to inspect the live process:

```
jcmd <pid> VM.info
```

2. Search for the `container` section in the output (cgroup). This provides a real-time snapshot of the container metrics, including `memory_usage_in_bytes`. Use this to determine if your heap size (typically a percentage of the limit) is appropriately leaving enough headroom for off-heap memory.

Troubleshooting Resource Constraints in OpenShift

When Java applications run in OpenShift, they operate within the constraints defined by the Linux kernel's control groups (cgroups). If these constraints are misconfigured or if the Java Virtual Machine (JVM) is unaware of them, you may encounter performance degradation, unexpected `OutOfMemoryError` (OOM) exceptions (that may, or may not exit the container), or even container restarts due to OOMKills.

This section focuses on diagnostic strategies and using tools to inspect resource constraints for Java workloads in OpenShift.

Diagnostic Strategy

To effectively troubleshoot resource constraints, you must correlate three distinct layers:

1. **OpenShift/Kubernetes Layer:** The resource requests and limits defined in your `Deployment` or `Pod` specification.
2. **Kernel/Cgroup Layer:** The actual hardware quotas enforced by the container runtime.
3. **JVM Layer:** The memory and CPU settings perceived by the Java process.

Identifying Misalignment

A common pitfall is the misalignment between the container limit and the JVM heap size. If your container limit is 1Gi but the JVM is configured with `-Xmx2G` (which should be avoided), the process will likely fail immediately or fail shortly after starting, being killed by the kernel when it attempts to allocate memory beyond the cgroup boundary.

Tools for verification

Using `oc describe`

The first step in any troubleshooting process is verifying the resource configuration of the pod.

```
oc describe pod <pod-name>
```

- **Look for:** The **Limits** and **Requests** sections.
- **Action:** Ensure the **Limits** are explicitly defined. If **Limits** are missing, the JVM may default to host-level visibility rather than container-level, potentially leading to incorrect heap sizing.

Leveraging **oc debug** and **inspect**

When an application behaves unexpectedly, you can use **oc debug** to inspect the runtime environment from inside the pod's namespace.

```
# Access the pod environment
oc debug pod/<pod-name> --image=registry.access.redhat.com/ubi9/ubi-minimal

# Once inside, inspect the cgroup limits directly
cat /sys/fs/cgroup/memory/memory.limit_in_bytes
```

Note: In **cgroup v2** environments, the path may be located under **/sys/fs/cgroup/memory.max**.

Using **jcmd**

The **jcmd** utility is the standard diagnostic tool for Java. Since Java 8u191+, the JVM is container-aware. You can use **jcmd** to verify how the JVM "sees" the container.

```
# Run jcmd inside the container to see VM flags
oc exec <pod-name> -- jcmd 1 VM.flags
```

- **Look for:** **MaxHeapSize**
- **Verification:** If **MaxHeapSize** is disproportionately large compared to your container limit, your application is at risk of being killed by the kernel.

Hands-on Lab: Troubleshooting a Resource-Constrained Pod

In this exercise, you will diagnose a pod experiencing memory constraints.

Prerequisites:

- An active OpenShift cluster.
- The **oc** CLI tool authenticated to your cluster.

Steps:

1. **Simulate a kill:** Deploy a pod with a very low memory limit (e.g., 64Mi) and a JVM heap requirement that exceeds directly set via **Xmx**. Observe the kill:

```
[source,bash]
----
oc get pods
# Look for OOMKilled status
```

```
----
```

2. Inspect the exit code:

```
[source,bash]
----
oc get pod <pod-name> -o
jsonpath='{.status.containerStatuses[0].state.terminated.reason}'
----
```

If the result is `OOMKilled`, the kernel terminated the process because it exceeded the cgroup memory limit.

3. **Verify the limits:** Check the deployment configuration to see the discrepancy between your **limit** and the JVM's **-Xmx** setting.
4. **Remediate:** Adjust the deployment YAML to either increase the container memory limit or update the JVM flags using the **JAVA_OPTS** (or **JAVA_OPTS_APPEND**) environment variables to reflect a lower heap percentage - both are available in run-java.sh script ENTRYPOINT.

```
[source,bash]
----
env:
  - name: JAVA_OPTS
    value: "-XX:MaxRAMPercentage=50.0"
----
```

5. **Re-verify:** After applying the change, use **jcmd** inside the container to confirm the new **MaxHeapSize** is calculated correctly based on the updated container limit.

Appendix

FIXME: Content for Appendix...

Complementary materials

Date	Title	Link
Mar 14, 2024	How to use Java container awareness in OpenShift 4	Read Article
May 16, 2024	Using OpenShift 4's inspect for middleware troubleshooting	Read Article
Mar 05, 2025	How to troubleshoot Spring Boot in OpenShift	Read Article
Nov 27, 2025	How does cgroups v2 impact Java, .NET, and Node.js in OpenShift 4?	Read Article
Jun 18, 2026	Troubleshoot application misbehavior after an OpenShift upgrade	Read Article

Resources

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